

概率论与数理统计 2022 Fall 半期 (回忆版)

Part One - Single Choice

1. Suppose events A and B are mutually exclusive, then which of the following is right?

A. $P(\bar{A}\bar{B}) = 0$

B. $P(AB) = P(A)P(B)$

C. $P(A) = 1 - P(B)$

D. $P(\bar{A} \cup \bar{B}) = 1$

2. Given $P(B) = 0.7$, $P(A|B) = 0.4$ and $P(A \cup B) = 0.8$, which of the following is false?

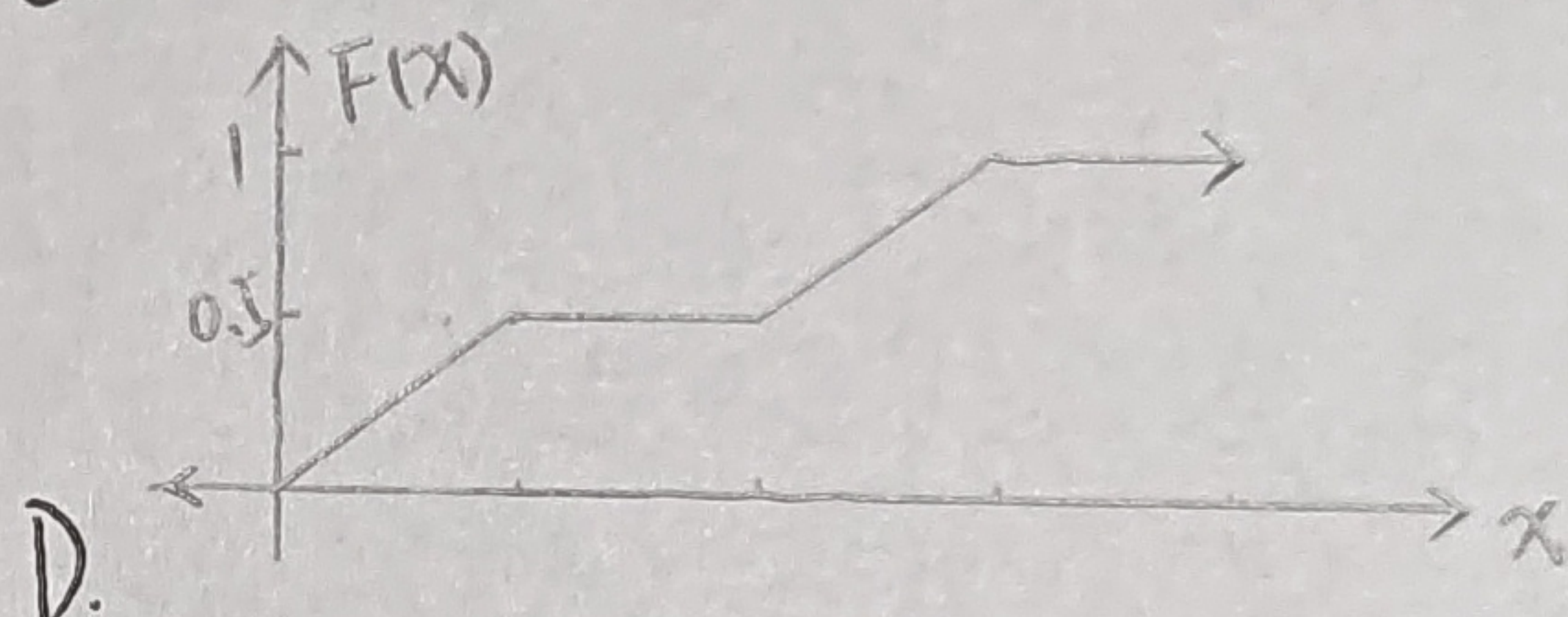
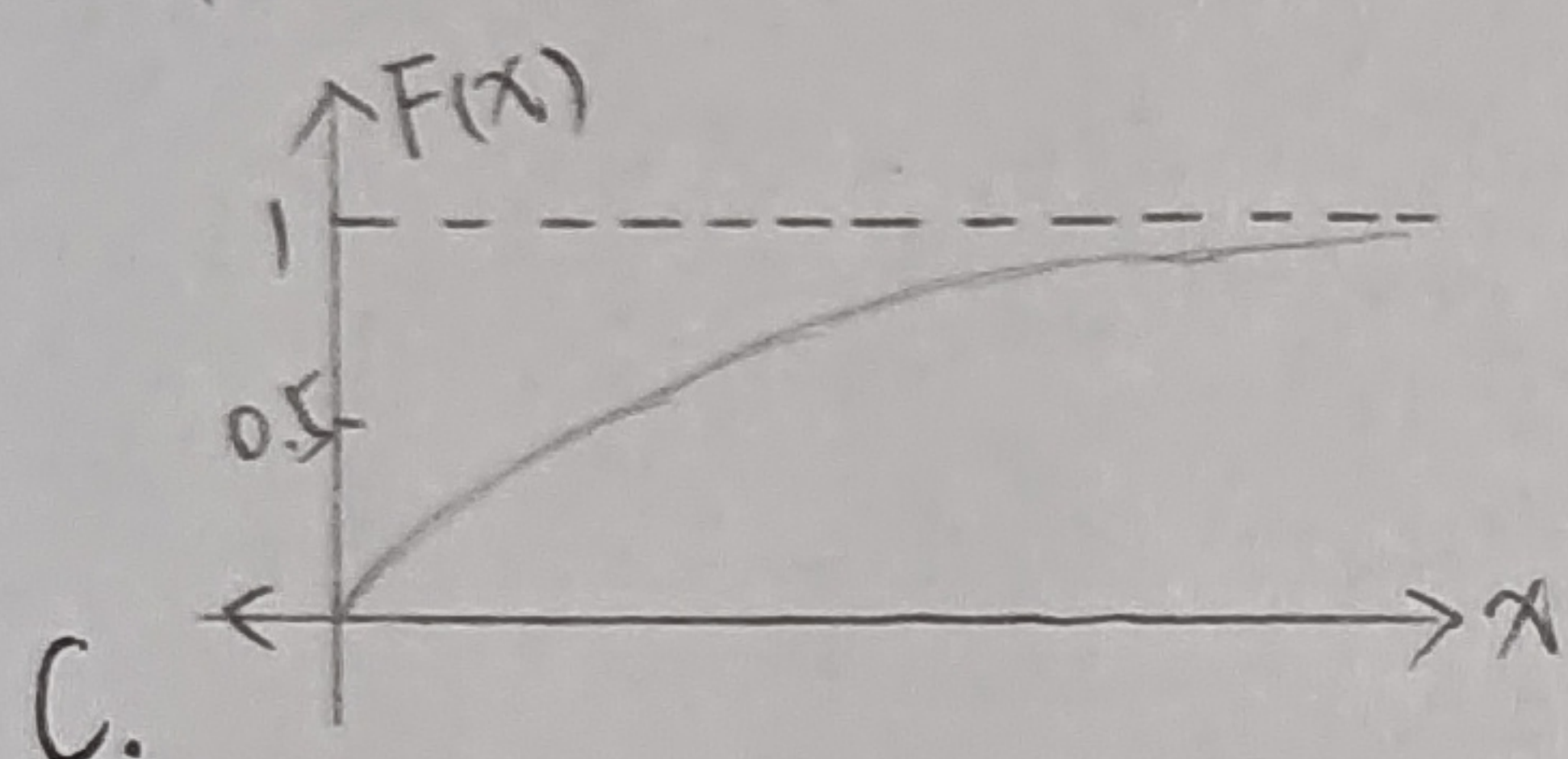
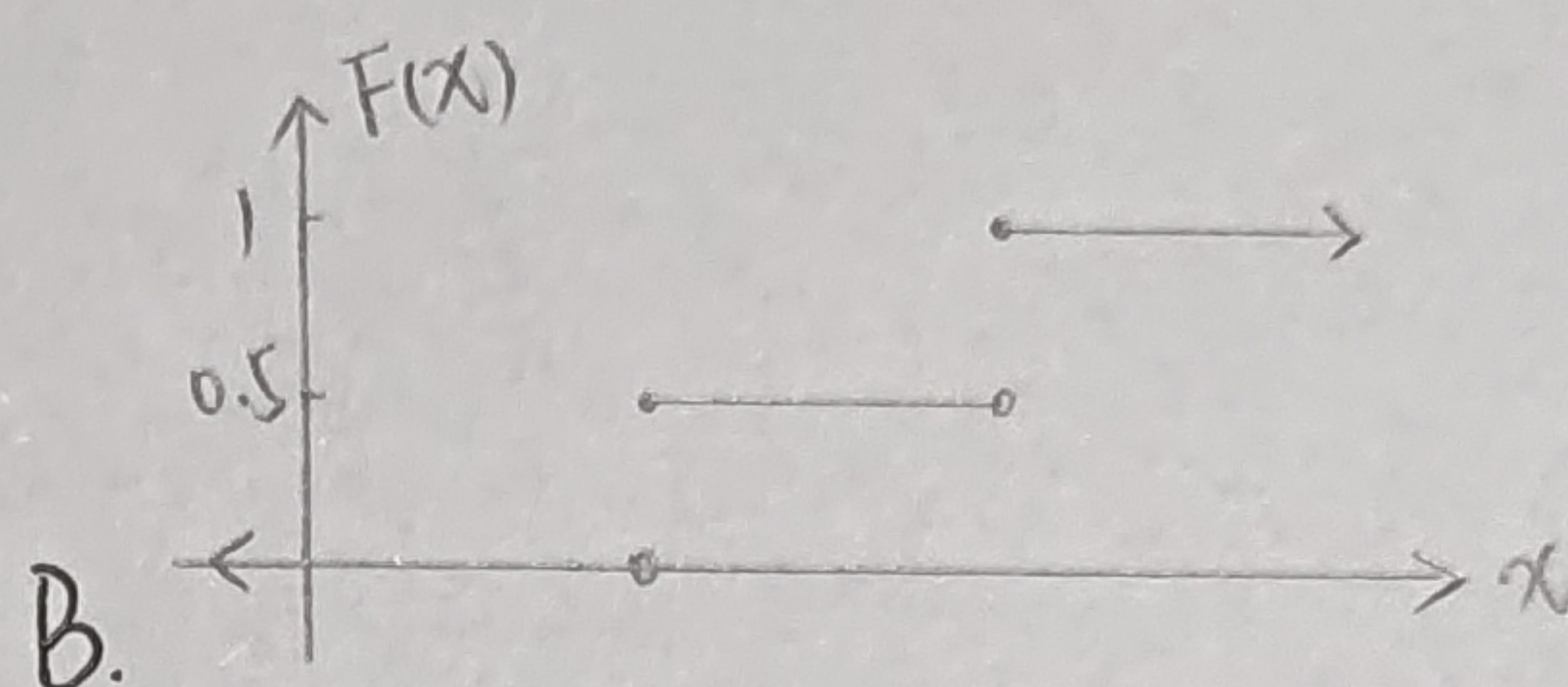
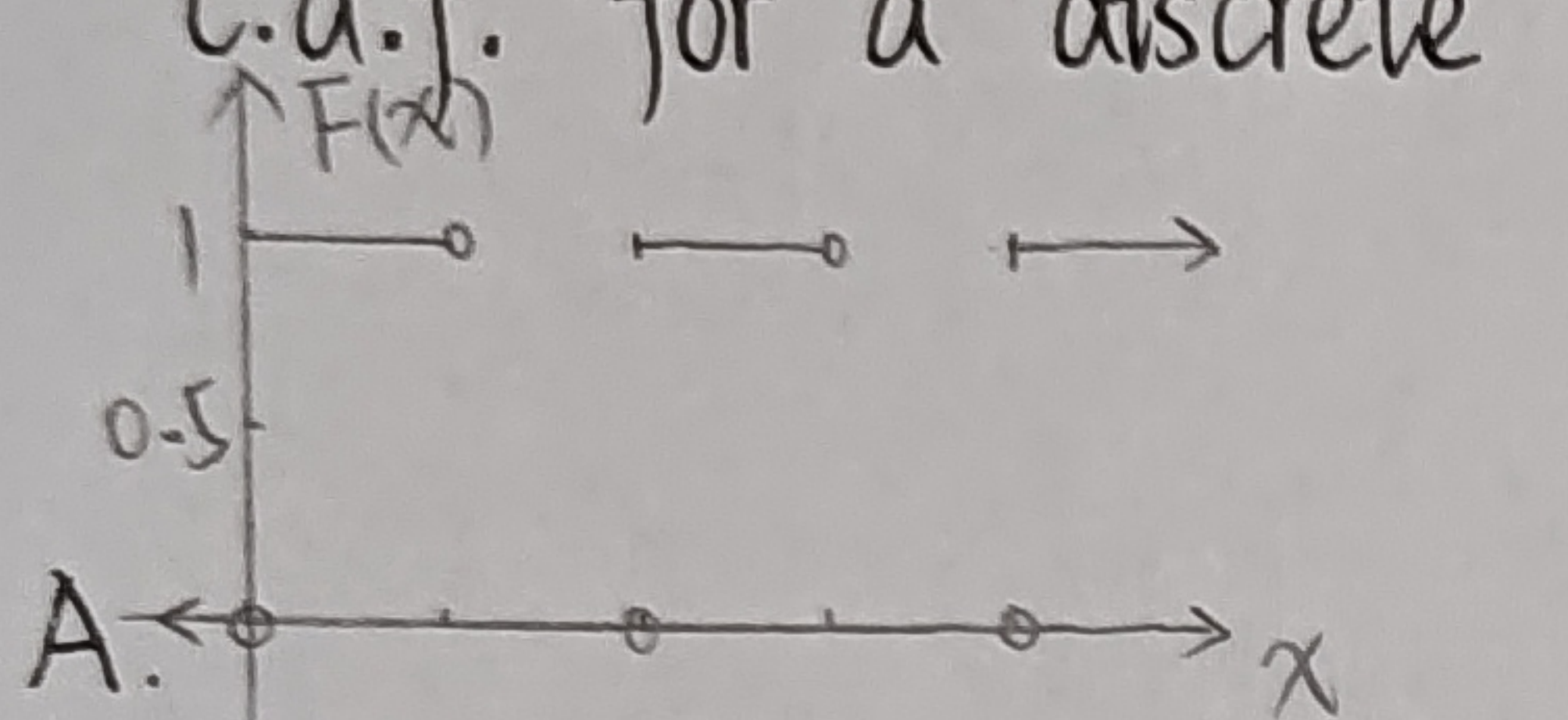
A. $P(A \cap B) = 0.28$

B. $P(A) = 0.38$

C. $P(AB) > 0$

D. $P(AB) = P(A)P(B)$

3. From the four given diagrams below, select the diagram that shows a valid c.d.f. for a discrete random variable.



4. Suppose the r.v. $X \sim U(2, 5)$. Select three independent observations from the distribution. What is the probability that at least two observations are less than 4?

A. $\frac{20}{27}$

B. $\frac{12}{27}$

C. $\frac{2}{5}$

D. $\frac{2}{3}$

5. The joint distribution function of random variables X and Y is given by:

$$P(X=x, Y=y) = \frac{x+y}{30}, \quad x=0,1,2,3; y=0,1,2.$$

What is the value of $P(X > Y)$?

A. $\frac{1}{2}$

B. $\frac{1}{5}$

C. $\frac{3}{5}$

D. $\frac{4}{15}$

Part Two - Blank Filling

1. Randomly put 3 books into 4 drawers. What is the probability that they are in the same drawer?

2. There are 3 white balls and several black balls of the same size in a bag, and the balls are touched 3 times. If the probability of touching at least one white ball is $\frac{19}{27}$, what is the number of black balls in the bag?

3. Given $P(A) = \frac{1}{4}$, $P(B|A) = \frac{1}{3}$, $P(A|B) = \frac{1}{2}$. What is the value of $P(A \cup B)$?

4. Let the random variable X follow a Poisson distribution with the parameter λ , and let $P\{X=4\} = P\{X=3\}$, then what is the value of λ ?

5. Let the r.v. $X \sim U(0, b)$ and $P\{1 < X < 3\} = \frac{1}{3}$. What is the value of b ?

6. Let r.v. $X \sim N(0,1)$. What is the distribution of $Y = -3X + 2$?
7. Suppose the discrete random variables X and Y are independent, where $P\{X=k\} = P\{Y=k\} = \frac{k+1}{3}$ for $k=0,1$. What is the value of $P\{X=Y\}$?
8. Suppose the joint density function of the continuous random variables (X,Y) is $f(x,y) = \begin{cases} Ae^{-(x+y)}, & x>0, y>0 \\ 0, & \text{others} \end{cases}$. What is the value of $P\{X < Y\}$?
9. Suppose the plane area D is surrounded by the curve $y = \frac{1}{x}$ and the straight lines $y=0, x=1, x=e^2$. The two-dimensional random variables (X,Y) are in D and follow a uniform distribution. What is the conditional density $f_{Y|X}(0.25|2)$?
10. Suppose two dimensional random variable $(X,Y) \sim N(1,0; 1,1; 0)$, then what is the value of $P\{XY - Y < 0\}$?

Part Three - Question Answering

1. A student takes two consecutive exams for the same course. The probability of passing the first exam is p . If the student passes for the first exam, the probability of passing the second exam is p too; if the student fails the first exam, the probability of passing for the second time is $\frac{p}{2}$.
 - (1) If the student passes the exam at least once, then he can obtain the qualification. Find the probability that he will obtain the qualification;
 - (2) If it is known that the student passed for the second exam, find the probability that he passed the first exam.
2. There are two boxes, the first box contains 2 red balls and 1 white balls, and the second box contains half red balls and half white balls. Take a ball from each of the two boxes and put them together, then take a ball from it.
 - (1) Find the probability that the ball is a red ball;
 - (2) If the ball is found to be a red ball, find the probability that the ball drawn from the first box is red.
3. A room has 3 windows of the same size, and only one of them is open. A bird flew into the room from the open window, it can only fly out from the open window. Birds are flying around the house trying to get out of the room. Assuming the bird has no memory, it flies to the windows randomly.
 - (1) Denote by X the number of times that the birds has tried in order to fly out of the room, find the frequency function of X ;
 - (2) The owner claims that a bird he keeps has a memory and that it does not

attempt to fly to any window more than once. Let Y represent the number of times that this clever bird has tried to fly out of the room. As the owner said, it is indeed, try to find the frequency function of Y ;

(3) Find the probability that the number of test flights X is less than Y .

4. Let the density function of X be $f(x) = \begin{cases} kx^2, & 0 \leq x < 3, \\ 0, & \text{others,} \end{cases}$. Let $Y = \begin{cases} 2, & X \leq 1, \\ X, & 1 < X < 2, \\ 1, & X \geq 2. \end{cases}$

(1) Find the constant k ;

(2) Find the distribution of function of Y ;

(3) Find the probability of $P\{X \leq Y\}$.

5. Suppose that the discrete random variables X and Y are independent, where X takes values from $\{0, 1, 2\}$, and Y takes values from $\{0, 1\}$. Moreover, $P\{X=0\} = \frac{1}{6}$, $P\{X=0, Y=1\} = P\{X=1, Y=0\} = \frac{1}{8}$.

(1) Find the joint frequency function of X and Y ;

(2) Find the frequency function of $X+Y$.

6. Let the joint density function of (X, Y) be $f(x, y) = \begin{cases} Ae^{-(x+2y)}, & 0 < y < x, \\ 0, & \text{others.} \end{cases}$

(1) Find the constant A ;

(2) Find the conditional density function $f_{Y|X}(y|x)$;

(3) Find the conditional distribution function $F_{X|Y}(x|y)$;

(4) Are X and Y independent? Explain the reason.